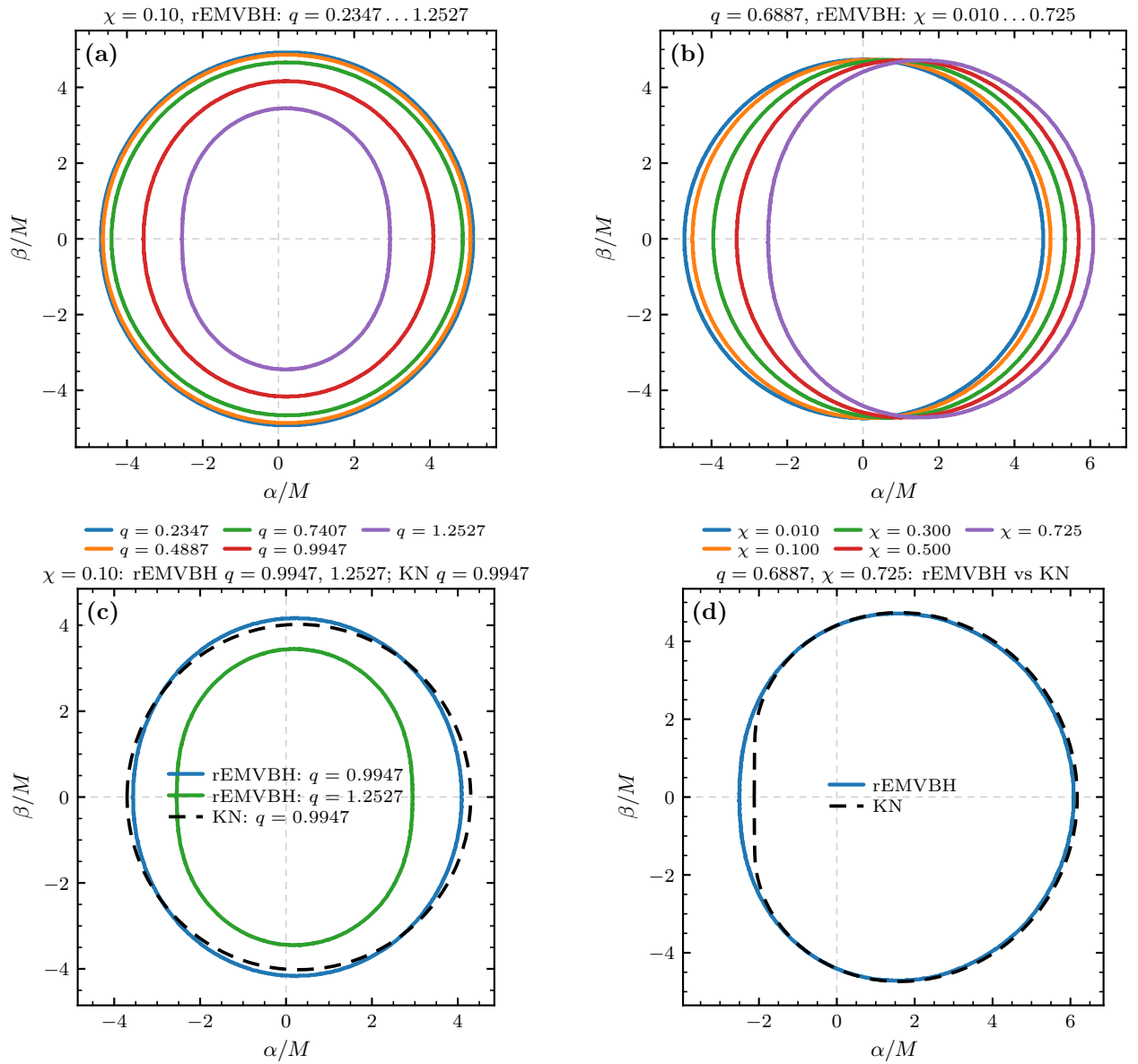


Solid curves: rEMVBH shadow, FOORT ray tracing in the observer's local orthonormal frame at $r_{\rm obs} = 1000\,M$.

Dashed curves: analytic Kerr–Newman shadow, asymptotic Bardeen impact parameters (α, β).

Pipeline offset (FOORT native KN vs analytic KN): $|\Delta W/W|$, $|\Delta H/H| \lesssim 0.1\%$, contour $\lesssim 0.02\,M$.